A PLANETARY-TO-NANOSCALE REASSESSMENT OF THE SUSPECTED BRUSHY CREEK IMPACT CRATER IN LOUISIANA. Dewan Mohammad Enamul Haque[1*], and Suniti Karunatillake[1]. [1]Department of Geology & Geophysics, Louisiana State University (* Email: dhaque1@lsu.edu) (E-mail: sunitiw@lsu.edu)

Introduction: Over the last three decades, scientists have been working to determine the origin of the Brushy Creek Feature (BCF), first documented in 1997 [1]. [2] in 2003, first hypothesized that BCF might form from an impact mechanism due to its topographically anomalous circular depression shape (FIG.1) and location in an area with no volcanic activity, salt diapirism, and carbonate sinkholes. This study also found shocked and fractured quartz, as supported by recent studies, e.g., [3, 4], which additionally observed planar deformation features (PDF) in fractured quartz. Since then, investigations of this suspected crater, located in St. Helen Parish, Louisiana (south of Route 37), have expanded beyond topo- & petrographic analysis; it has also been investigated geophysically. Although [5] observed stratigraphic discontinuity and high resistivity anomaly, respectively from the GPR and resistivity survey transecting the BCF, the Bouguer anomaly was less than the expected 1-10mGal negative value [6]. These gravity survey-based results do not support the impact origin of Brushy Creek.

We sought to address this interesting research problem differently, using landscape evolution modeling and connecting additional evidence through microanalysis, such as XRD and Laser Raman Spectroscopy.

Methods: We have performed landscape evolution modeling (LEM) using Landlab [7]. We simulated topographic evolution from 35 ka to the present day, combining flow routing with stream-power incision and linear hillslope diffusion to represent soil-mantled surface processes. Starting from a simplified, smoothed pre-impact surface, we emplaced parametric crater geometries of varying diameter and depth at candidate impact times. Subsequently, we evolved the landscape forward to the present day. We evaluated model results by comparing modeled and observed present-day topography using RMSE calculated outside the crater zone and against a no-impact baseline to assess whether including an impact feature improved the fit.

Following recommendations from previous studies [3,4], we re-conducted microanalytical investigations to obtain additional petrographic evidence using reflection- and transmission-based techniques (optical microscopy and cathodoluminescence), scattering-based methods (electron microprobe and Raman spectroscopy), and diffraction-based analyses (XRD and XRF).

Results & Interpretations: Our performed LEM [https://doi.org/10.5281/zenodo.18190400] indicates that a shallow (5-meter), kilometer-scale (1.6km) impact-like structure (FIG.2) might have formed during the late Pleistocene (with older ages in the tested window—around ~35 ka BP, mostly favored), which is consistent with, and modestly improves the fit (FIG.3) to the present-day topography of the BCF area. While non-impact processes cannot be ruled out, the systematic reduction in misfit (FIG. 4A & 4B) when an impact-like feature is included suggests that an impact origin remains a plausible explanation. Because the misfit improvement varies only weakly (FIG. 4A & 4B) with age across the tested range, the impact timing is not tightly constrained, and we interpret broadly as a possible late Pleistocene origin rather than a single definitive date.

We carried out a detailed microanalysis (methods) of Brushy Creek samples; they are sedimentary, primarily composed of quartz, feldspar, zircon, garnet, apatite, rutile, and minor sulfur & ilmenite (FIG.6 & XRF). We find that quartz is highly fractured (FIG. 7), as reported in previous studies. The quartz also shows varying levels of luminescence and PDF (FIG. 7). Raman spectra (FIG. 5) confirm the presence of quartz & chalcedony, but no high-pressure quartz varieties, such as tridymite, coesite, or cristobalite, are found, unlike other crater sites [e.g., 8]. Further research is still required as our study depended on samples from only one core & this sample was collected from the center of the suspected crater site. Analyzing more samples from the rim & central suspected crater & beyond is required to solve this dilemma.

Conclusions & Implications: It is quite challenging to determine the origin of Brushy Creek's impact from LEM because it lies in erosion-prone sedimentary and subtropical settings. Still, the LEM method that we employed retrieves a promising resolving signal. Future studies may build on indications from forward LEM results (highlighting the possible crater rim and suspected center) (FiG.1,2,3) to collect further representative samples for micro- and pollen analysis. Raman Spectroscopy may be used to identify high-pressure quartz polymorphs. U–Pb geochronology of zircon, titanite, rutile, and apatite may provide robust age constraints, particularly through the identification of impact-generated melt and newly crystallized phases. Pollen analysis may help determine whether there is any impact-driven ecological disruption.

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References: [1] McCulloh et al. (1997) *USGS Report agreement no. 1434-HQ-96-AG-1490.* [2] Heinrich P. (2023) *LGS report#50006*. [3] Schedl A. (2018) LGS, Open-File Series No. 18-01. [4] Matherne H. (2020) LPSC #2361. [5] Hood D. (2020) LPSC #2215. [6] James P. (2018) LPSC #1512. [7] Barnhart et al. (2020) https://doi.org/10.5194/esurf-8-379-2020. [8] Chandran et al. (2021) https://doi.org/10.1016/j.lithos.2021.106479. [9] USGS (2021) https://doi.org/10.5069/G98K778D.

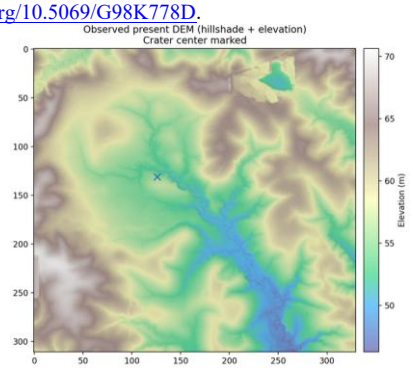


FIG.1 Present-day topography [9]

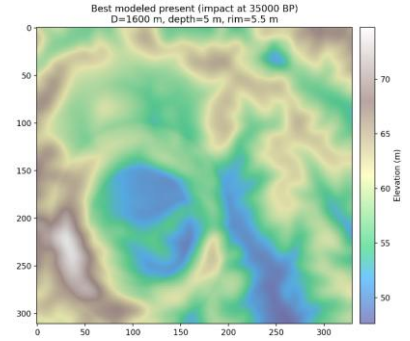


FIG.2 Modeled present-day topography

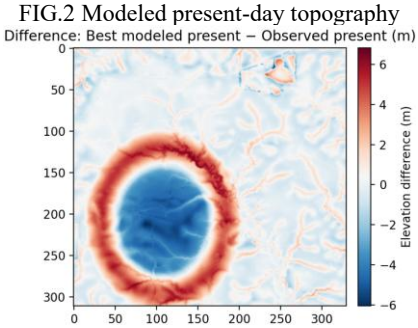
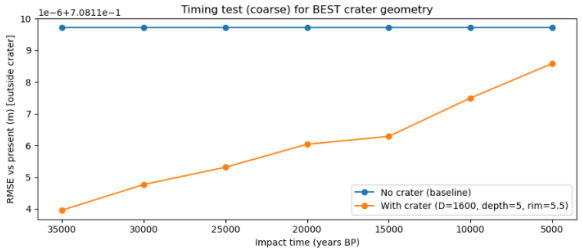


FIG.3 Topography difference



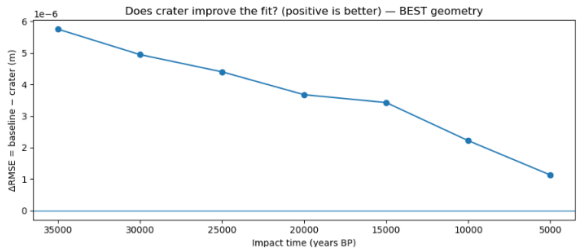


FIG. 4A & 4B RMSE (outside crater)

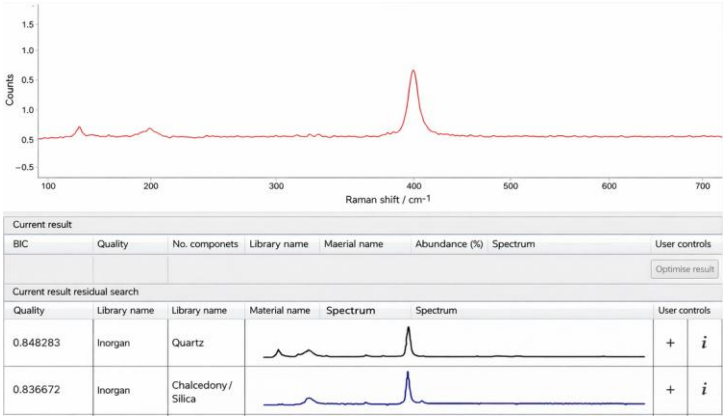


FIG.5 Raman Spectra

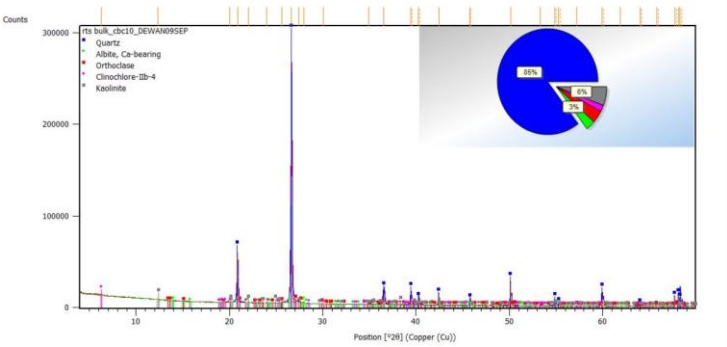


FIG. 6 XRD Analysis

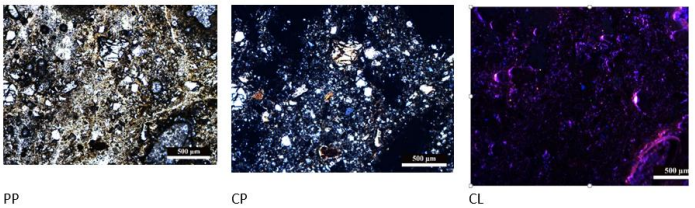


FIG.7 Optical microscope analysis and cathodoluminescence